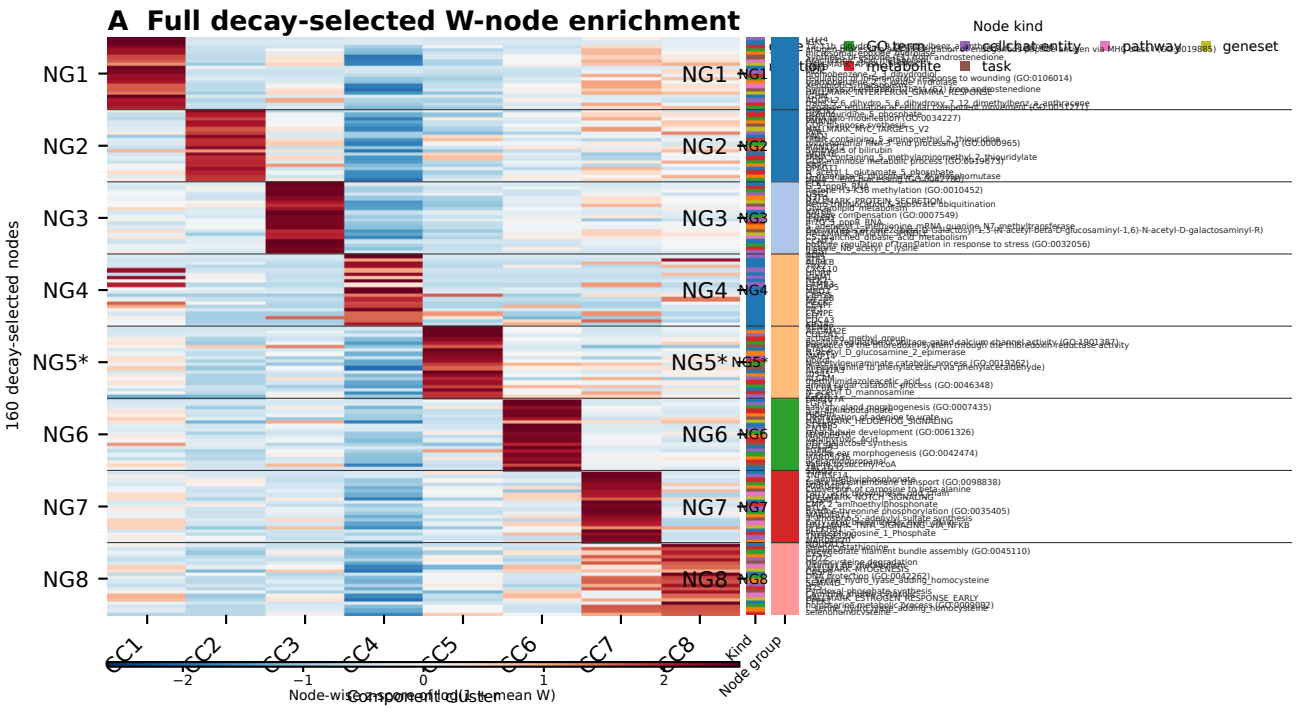
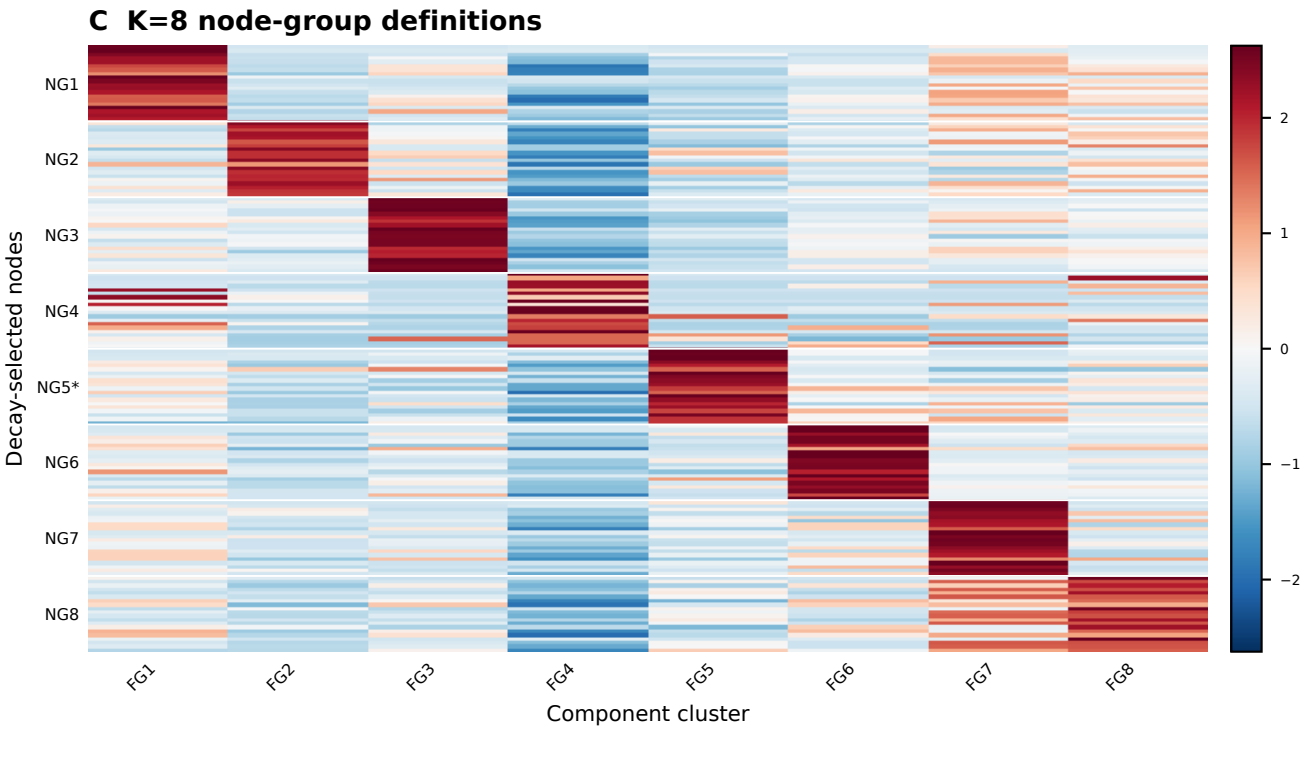
TNBC epithelial H→node groups after annotation-node exclusion

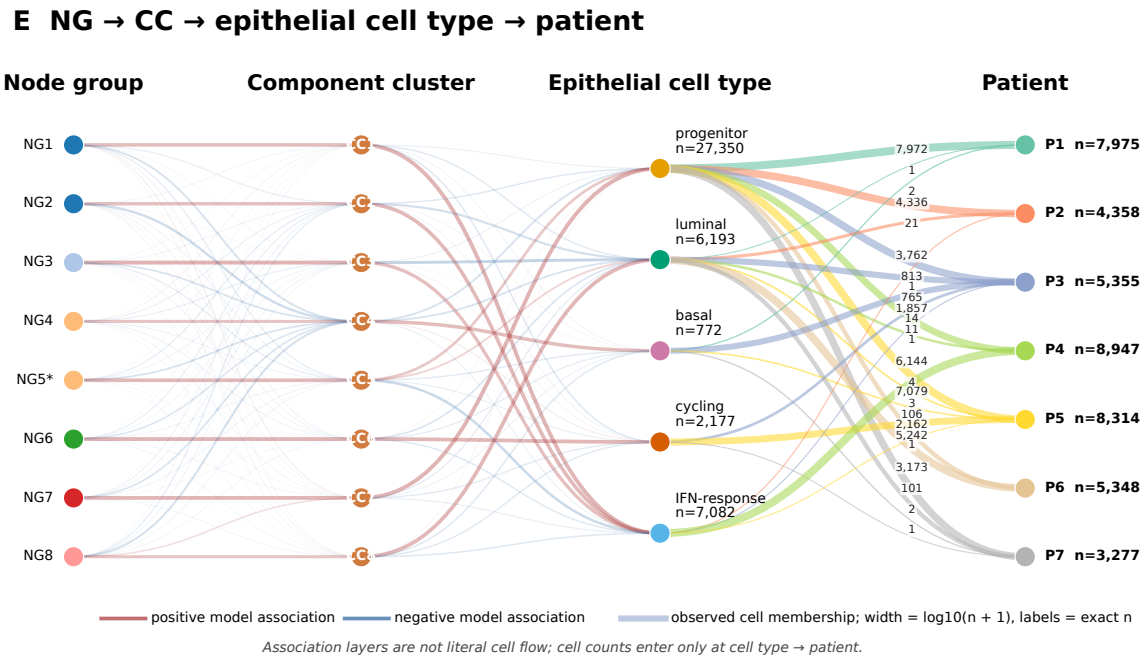
A Full decay-selected W-node enrichment



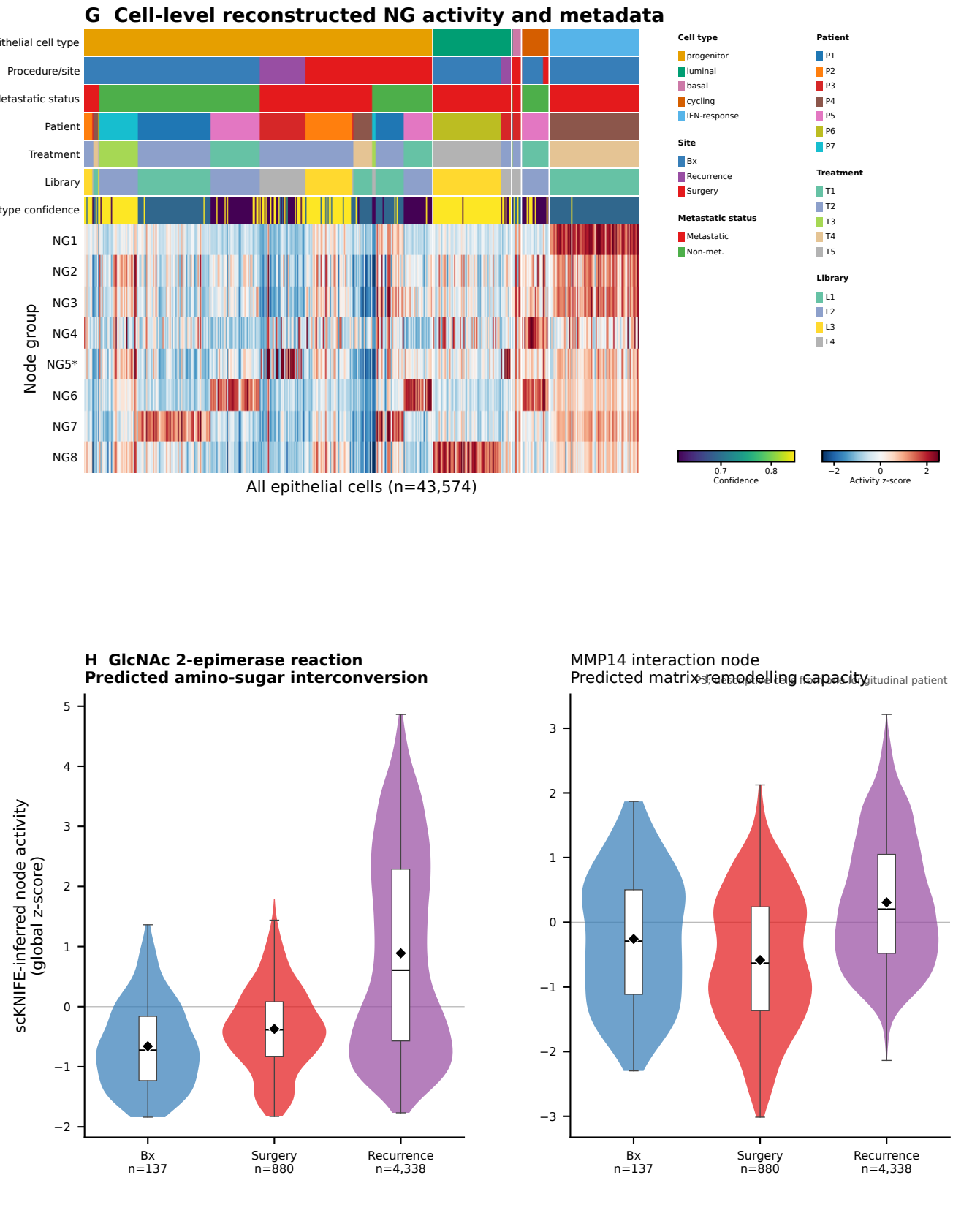
C K=8 node-group definitions



E NG → CC → epithelial cell type → patient



G Cell-level reconstructed NG activity and metadata



H GlcNAc 2-epimerase reaction
Predicted amino-sugar interconversion

MMP14 interaction node
Predicted matrix-remodelling capacity

Cell-type and cell-state graph nodes are excluded before W clustering; epithelial cell types remain biological activity groupings.